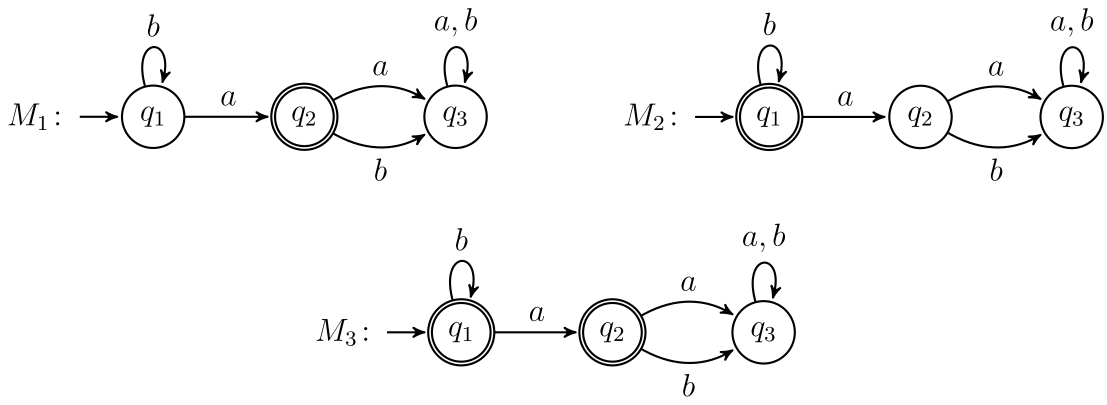
Exercise 1

Let $L_1 = \{aa, bb, bbb\}$ and $L_2 = \{abba, aab, bb\}$ be two languages over the alphabet $\Sigma = \{a, b\}$. What are the strings of the following languages?

1. $L_1 \circ L_2$
2. $L_1 \cup L_2$
3. $L_1 \cap L_2$
4. Provide a few strings of L_2^*

Exercise 2

Given the following three automata M_1, M_2, and M_3



1. Describe the sequence of states that M_1 goes through while reading the inputs:
   (a) $abbbab$
   (b) $ababaab$
   (c) $aaaaa$
   (d) ε
2. Which of the previous sequences are accepting in M_1, M_2 and M_3?
3. Describe the languages recognised by each of the three automata.

Exercise 3*

Provide the state diagram of the automaton $M_4 = (Q, \Sigma, \delta, q_0, F)$ given below and describe its language.

$$Q = \{q_0, q_1, q_2, r_1, r_2\}$$
$$\Sigma = \{a, b\}$$
$$F = \{q_1, r_1\}$$

δ	a	b
q_0	q_1	r_1
q_1	q_1	q_2
q_2	q_1	q_2
r_1	r_2	r_1
r_2	r_2	r_1

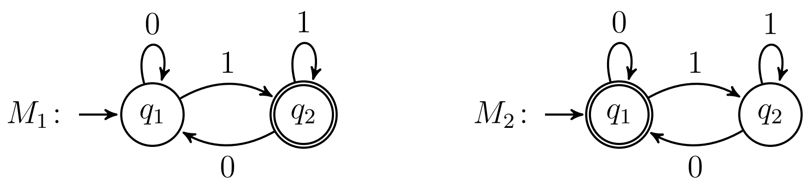
Exercise 4*

For each of the following languages, draw the state diagram of an automaton recognizing it.

1. $L_1 = \{w \in \{1, 22\}^* \mid w \text{ has } 11 \text{ as prefix}\}$
2. $L_2 = \emptyset \subseteq \{0, 1, 2\}^*$
3. $L_3 = \{\varepsilon\} \subseteq \{0, 1, 2\}^*$
4. $L_4 = \{w \in \{\texttt{go}, \texttt{stop}\}^* \mid w = \varepsilon \text{ or ends with } \texttt{stop}\}$
5. $L_5 = \{w \in \{0, 1\}^* \mid w \text{ has } 001 \text{ as prefix or } 11 \text{ as suffix}\}$

Exercise 5*

Consider the automata M_1 and M_2 given below.



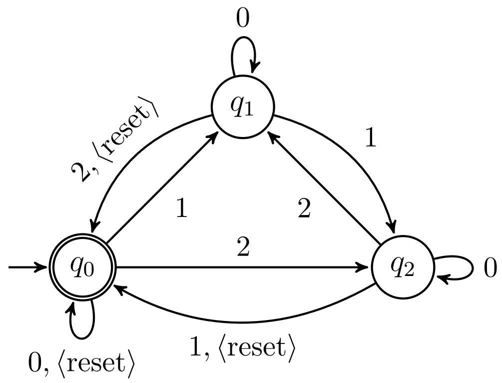
1. Construct an automaton that recognizes the language $L(M_1) \cap L(M_2)$;
2. Construct an automaton for each of the following languages:
   (a) $\overline{L(M_1)}$
   (b) $\overline{L(M_2)}$
   (c) $\overline{L(M_1) \cap L(M_2)}$

Exercise 6

1. Prove that regular languages are closed under intersection.
   (Hint: Similar construction to the one for the union, only that $F = F_1 \times F_2$);
2. Prove that regular languages are closed under complement.
   (Hint: Transitions don't change. But, what should be now the accepting states?).

Exercise 7*

Let M be the following automaton over the alphabet $\Sigma = \{0, 1, 2, \langle\text{reset}\rangle\}$.



1. Give examples of accepted and non-accepted words (at least five for each).
2. Define $count(w)$ to be the sum of the numerical symbols in a string $w \in \Sigma^*$. Prove that $L(M) = \{w \mid (count(w)(\text{mod } 3)) = 0\}$.
   (Hint: Note that M keeps a running count of the sum of the numerical input symbols it reads, modulo 3. Every time it receives the ⟨reset⟩ symbol it resets the count to 0).

Exercise 8 (optional)

Generalize the automaton M from Exercise 7 so that $L(M) = \{w \mid (count(w) (\text{mod } 4)) = 0\}$.